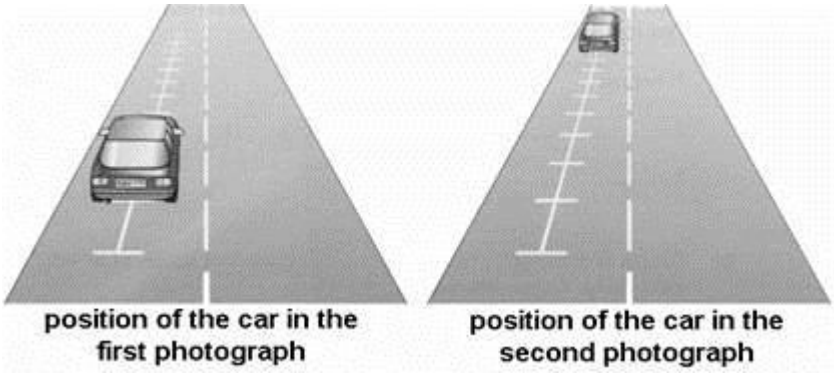
Q1.

Speed cameras are used to detect motorists who break the speed limit. A number of lines 2 m apart are painted on the road. As a speeding car crosses the painted lines, the camera takes two photographs, 0.5 s apart.



(a) (i) How far did the car move between the two photographs?
Give the correct unit.

.....

.....

1 mark

(ii) How fast is the car in the photographs moving?

.....

..... **m/s**

1 mark

(b) It takes 0.0002 s to take each photograph.
How far does the car move while the speed camera is taking **one** photograph?

.....

..... **m**

1 mark

(c) The speed camera gives out bright flashes to provide enough light for the photographs.
How does the light from the flash get back to the camera to produce the photographs?

.....

.....

1 mark

Maximum 4 marks

Q1.

(a) (i) 16 m

unit required for the mark

1

(ii) 32

consequential marking applies

accept answer to(a)(i) ÷ 0.5

1

(b) 0.0064

accept '6.4 mm' or '0.64 cm' consequential marking applies
accept answer to (a)(ii) × 0.0002 s

1

(c) it is scattered from the car **or** the road **or** the lines

*accept 'it is reflected' **or** 'it bounces off*

1

[4]